EGME-3170 Thermal Systems Group Project

Problem Statement and Instructions

Problem Statement: Nine CPUs (Intel Core i7 12700K) are mounted to the outer surface of each side of a square duct as shown in Figure 1. The duct has inside dimensions of 15-cm by 15-cm, duct material thickness of 1 cm, and length of 15 cm. Each CPU has a surface area of 0.0014 m^2 and is attached to the duct using a thermal paste, The sides of the duct act as a thermal spreader. For calculating fin efficiency and convective heat transfer in this project, you can assume the base temperature of the duct is uniform and constant.

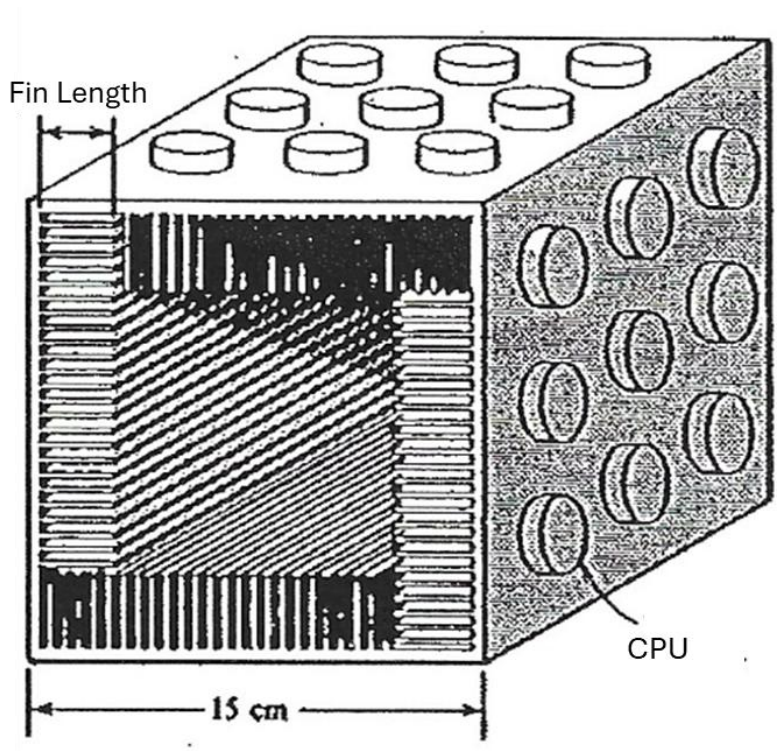

Figure 1. Duct configuration

The PassMark benchmark score is used as a surrogate for CPU performance, CPU_{perf}. This can be computed as a function of total power, P_{CPU} in W, using equation (1):

$$CPU_{perf} = 39{,}000\left(1 - exp(-0.02 * P_{CPU})\right) \quad (1)$$

You must design a fin bank for the inside surface of the duct, subject to the constraints in Table 1, while seeking to maximize CPU_{perf} / P_{Tot} where the total power is the sum of the CPUs and the fan used to cool the system. The fan flowrate, Q_{fan} in L/min, is a function of the fan power, P_{fan} in W, as given by equation (2):

$$Q_{fan} = 100{,}000\left(\frac{P_{fan}}{10{,}000W}\right)^{1/3} \quad (2)$$

Table 1. Project design constraints

Requirement #	Requirement Statement
1.1	The inner dimensions of the duct *shall* be 15-cm by 15-cm.
1.2	The total length of the duct *shall* be 15 cm
1.3	The duct wall thickness *shall* be 1 cm
2.1	Nine CPUs *shall* be attached to each side surface of the duct.
2.2	The temperature of each CPU *shall not* exceed 90 ℃
2.3	Thermal paste *shall* be used between each CPU and the duct
3.1	The cooling fluid *must* be air.
3.2	The inlet temperature of the cooling fluid *shall* be 20 ℃
3.3	The exit temperature of the cooling fluid *shall not* exceed 50 ℃
3.4	The fan power *shall not* exceed 1 kW
4.1	The duct and fin(s) *shall* be made of the same material
4.2	The duct and fin(s) *may* be copper, aluminum, or 316 stainless steel
4.3	The fin(s) *shall* be equally spaced across each surface as shown in Fig. 1
4.4	The fin(s) *shall* use a straight rectangular profile
4.5	The fin(s) *shall* extend the entire length of the duct.

Part 1: Develop a numerical model for the required fin bank. You must use Example 2.11 from the Thermal Conduction notes (slide 79) to verify the model.

a) Produce a diagram/drawing with dimensions showing the relative geometry of each fin.
b) Draw a thermal resistance diagram for the entire fin bank including thermal resistance associated with the finned and unfinned regions, the duct thickness, and the contact between the CPU and the duct.
c) List your assumptions for the fin bank design
d) Develop the numerical model in terms of the CPU power, fins per side, the length of each fin, and the thickness of each fin. All governing equations must be included in the report and formatted using an Equation Editor as shown in Eq. 1 in this report. You do not need to provide explicit forms of each equation, but you need to define each term including the units.
e) Verify the numerical model by solving Example 2.11(a) from the Thermal Conduction notes (slide 79). For this step you can neglect thermal resistance from the thermal paste and the duct thickness and assume the convection heat transfer coefficient given in the example problem.

Part 2: Add cooling flow to the numerical model. Calculate the fan power required to cool the transistors ("CPUs") for the geometry given in Example 2.11 from the Thermal Conduction notes (slide 79).

a) Produce a diagram/drawing showing the surrogate "duct" geometry between each fin. You can treat this as a fully enclosed duct for this problem.
b) Treating the area between the fins as a closed duct, compute the hydraulic diameter and Reynolds number for flow between each fin.

c) Estimate the bulk temperature in the duct using the Conservation of Energy and simple average method. The mean velocity can be computed using the Conservation of Mass.
d) Compute the convective heat transfer coefficient, h, for flow between the fins. You may assume that flow is turbulent within the duct.
e) Using the fin bank geometry and CPU power from Example 2.11 in the Thermal Conduction notes (as verified in Part 1), compute the fan power, P_{fan} in W, required to cool the system.

Part 3: Evaluate system performance as a function of fin bank design parameters. You should use a fin bank geometric design point as given in Example 2.11 in the notes. Specifically, 24 fins per side with dimensions of 15 cm wide, 2.5 cm long, and 2 mm thick.

a) At a single-CPU PassMark benchmark score of $CPU_{perf} = 30{,}000$, vary the duct material, number of fins, fin thickness, and fin length to maximize the performance metric, CPU_{perf} / P_{Tot}. This can be demonstrated by providing plots of the performance metric as a function of each design parameter varied around your fin design point.
b) Plot the performance metric, CPU_{perf} / P_{Tot}, as a function of the number of fins (1 to N) for a single-CPU power range of 10-190W at your fin design point.
c) Evaluate your assumptions from Part 1 and Part 2. Discuss their validity over the range of conditions.

EGME-3170 Thermal Systems Project: Report Format

0) **Cover Sheet:** List your name, course number, project title, and date. A brief summary of each student's contribution to the project should be included.
1) **Introduction:** Write a concise overview of the problem to be solved along with your proposed numerical method(s) for Parts 1–3.
2) **Assumptions:** A numbered list of your assumptions with a brief description of each assumption, why it is justified, and the effect that it has on your analysis. Make sure to evaluate the appropriateness of these assumptions based on your results.
3) **Calculations:** As requested in Parts 1–3, show how you set up the equations in your calculations for each step. You do not need to arrange the equations in explicit form but may use implicit forms. All equations must be included in the report and formatted using an Equation Editor as shown in Eq. 1 in this report.
4) **Results:** Provide your results in tables or figures as requested in Parts 1-3. You must also explicitly show that all requirements in Table 1 are satisfied. Make sure all tables and figures are numbered and titled (as shown by Figure 1 and Table 1). Ensure that you discuss the results shown in ***every*** table and plot. Data tables for each plot must be provided in the appendix.
5) **Conclusions:** Write a concise conclusion that summarizes the primary results.
6) **Appendices:** You must include data tables for every plot in Part 3a and 3b.

EGME-3170 Thermal Systems Project: Rubric

Description	Max Pts	Mark
Cover Sheet (name, course, title, date, one paragraph contribution statement for each group member)	**5**	
Introduction and Assumptions (10 pts)		
Concise problem statement and description of the method of solution.	**5**	
List of assumptions with justification or description	**5**	
Calculations and Verification (25 pts)		
Diagrams of fin bank showing fin geometry and flow areas	**5**	
Accuracy of equations and verification of Part 1	**10**	
Accuracy of equations and verification of Part 2	**10**	
Table describing compliance with every requirement	**5**	
Results (45 pts)		
Part 3a: Variation of duct material	**5**	
Part 3a: Variation of the number of fins	**10**	
Part 3a: Variation of fin thickness	**10**	
Part 3a: Variation of fin length	**10**	
Part 3b: Variation of CPU power and number of fins	**5**	
Evaluation of assumptions from Part 1 and 2 based on results	**5**	
Conclusions (one or two paragraphs is sufficient)	**5**	
Appendices (Data tables for any plots provided in report)	**5**	
Total :	**100**	